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PERFORMANCE ANALYSIS OF CONVOLUTIONAL NEURAL NETWORK ARCHITECTURES FOR IMAGE CLASSIFICATION

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CONTENT

CONTEXT & MOTIVATION

- **Problem**: Deploying CNNs in resource-constrained environments (Edge, Embedded, Shared GPU Clusters).
- **Trade-off**: High predictive performance vs. low latency and memory usage.
- **Goal**: Systematically benchmark lightweight architectures under a unified, reproducible pipeline.

Light Models (Green)

- MobileNet V1
- ShuffleNet V2
- SqueezeNet
- EfficientNet-B0

Medium Models (Blue)

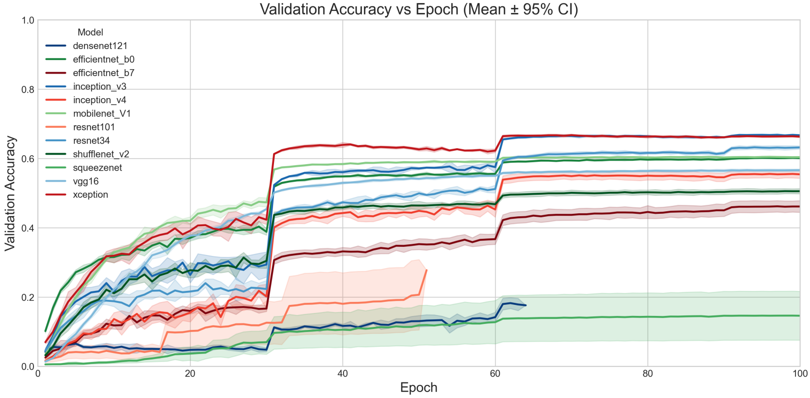
- VGG16
- ResNet34
- Inception V3
- DenseNet121

Heavy Models (Red)

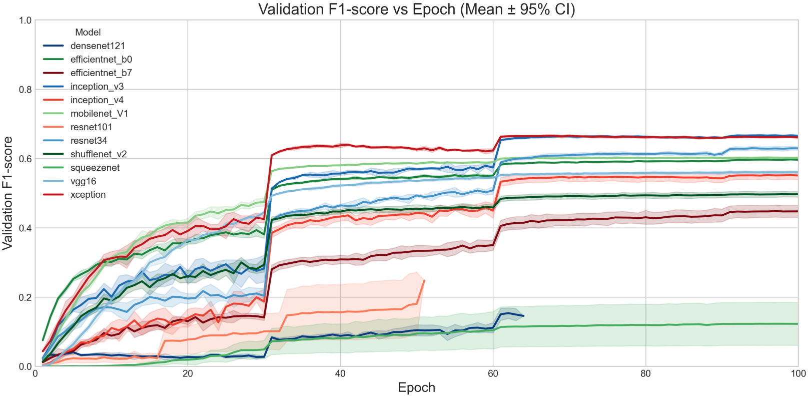
- ResNet101
- Inception V4
- Xception
- EfficientNet-B7

- **Hardware**: UFRGS PCAD Cluster (Tupi partition).
  - Single NVIDIA GPU (24 GiB VRAM) per job.
  - Data staged to node-local `SCRATCH` to minimize I/O latency.
- **Software**:
  - Python 3.11, PyTorch 2.0.
  - `PYTORCH_CUDA_ALLOC_CONF` tuned to reduce fragmentation.
- **Dataset**: Tiny ImageNet (200 classes, 64×64 px).
- **Protocol**: 100 Epochs, 5 Random Seeds per model.

VALIDATION ACCURACY VS. EPOCH



F1 SCORE VS. EPOCH

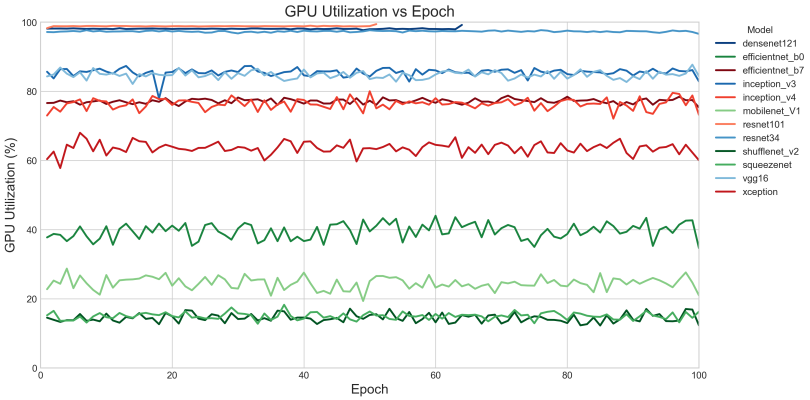


PEAK PERFORMANCE SUMMARY

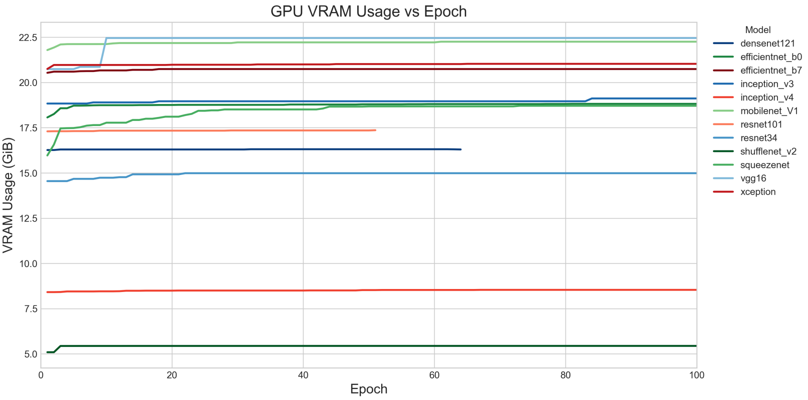
Model	**Top-1 Accuracy**	**F1-Score**
DenseNet121	0.184 ± 0.015	0.154 ± 0.015
EfficientNet-B0	0.603 ± 0.002	0.598 ± 0.002
EfficientNet-B7	0.463 ± 0.017	0.449 ± 0.018
Inception V3	0.670 ± 0.002	0.669 ± 0.002
Inception V4	0.557 ± 0.011	0.553 ± 0.011
MobileNet V1	0.606 ± 0.002	0.604 ± 0.002
ResNet101	0.207 ± 0.102	0.181 ± 0.092
ResNet34	0.633 ± 0.005	0.630 ± 0.005
ShuffleNet V2	0.506 ± 0.008	0.498 ± 0.008
SqueezeNet	0.147 ± 0.071	0.123 ± 0.062
VGG16	0.567 ± 0.004	0.561 ± 0.004
Xception	0.669 ± 0.003	0.667 ± 0.003

Tabela: Peak metrics (Mean $\pm$ 95% CI) across 5 seeds.

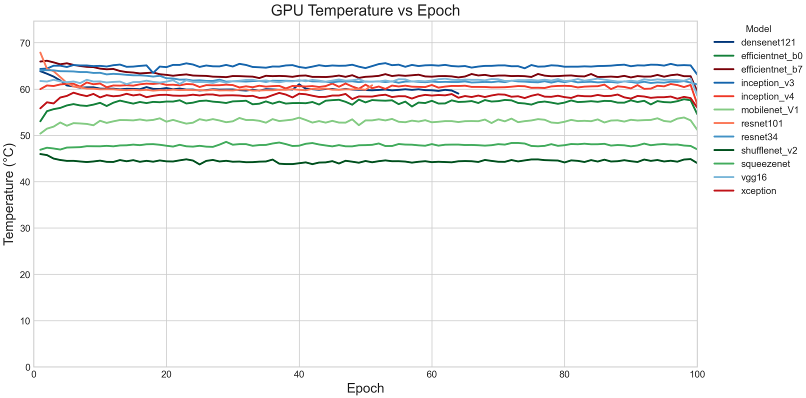
GPU UTILIZATION VS. EPOCH



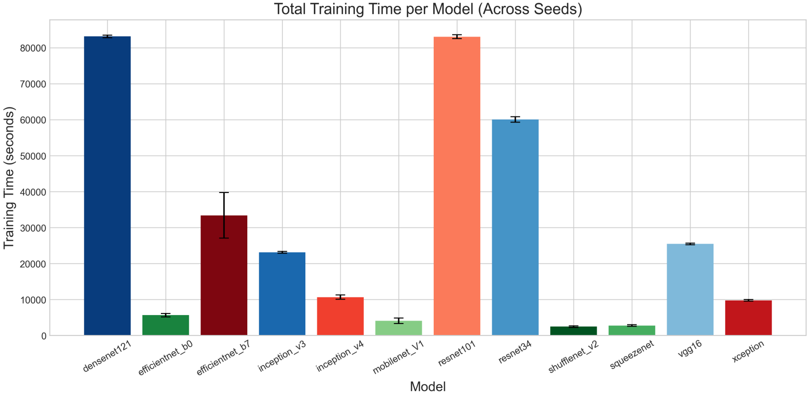
GPU VRAM USAGE VS. EPOCH



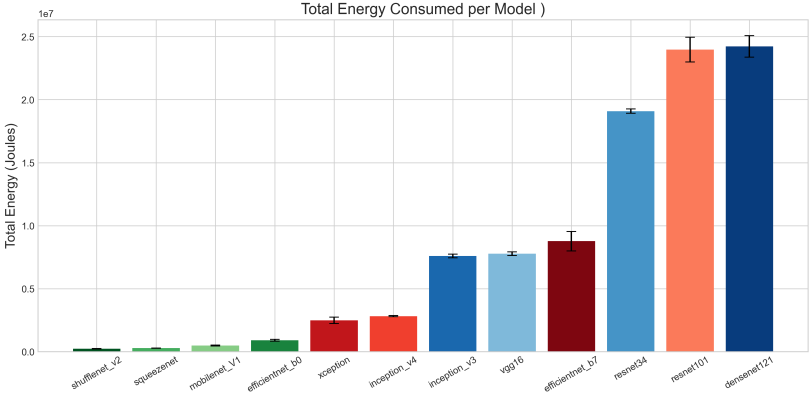
GPU TEMPERATURE VS. EPOCH



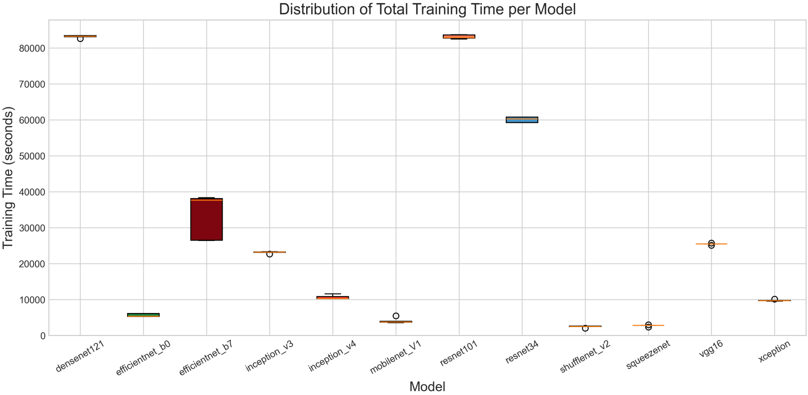
TOTAL TRAINING TIME



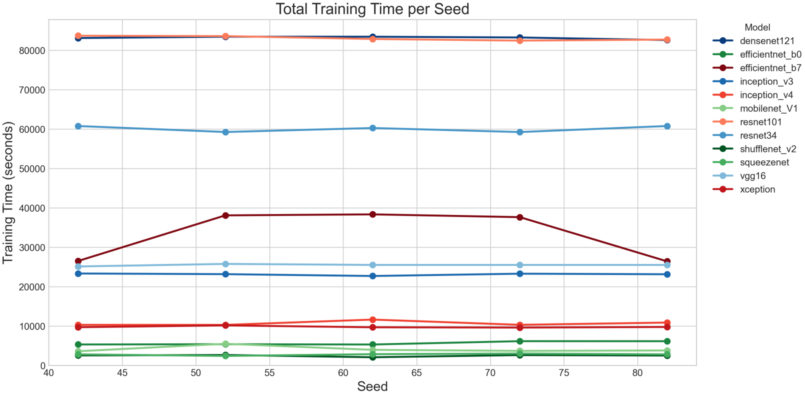
TOTAL ENERGY CONSUMPTION PER MODEL



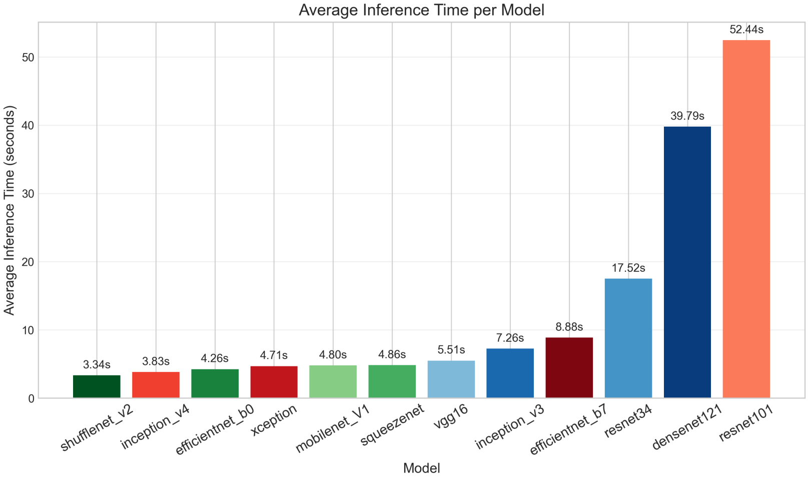
TRAINING TIME DISTRIBUTION



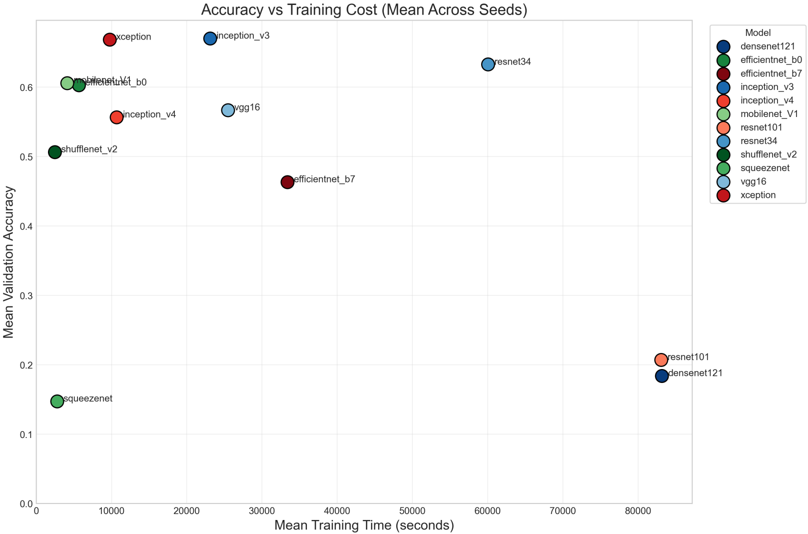
TRAINING TIME PER SEED



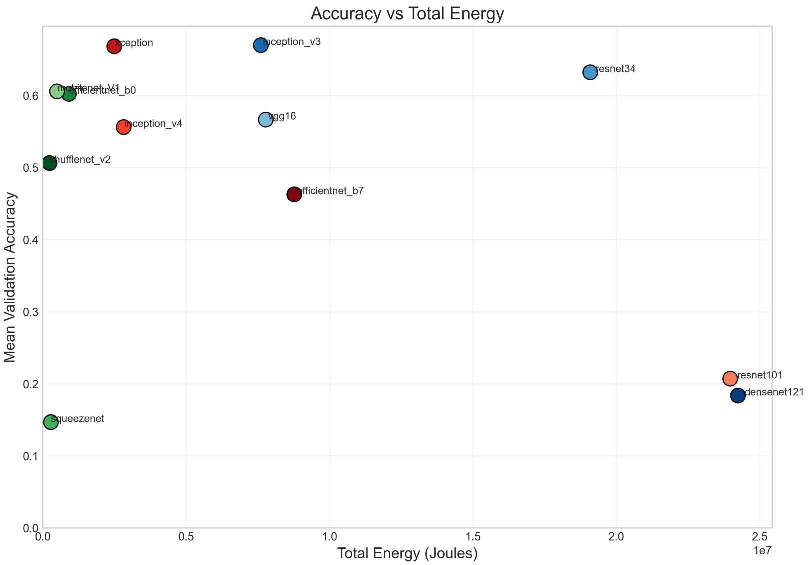
AVERAGE INFERENCE TIME PER MODEL



ACCURACY VS. TRAINING COST



ACCURACY VS. ENERGY CONSUMPTION



1. **Performance does not strictly follow model size.** Surprisingly, several heavy models underperformed compared to lightweight architectures. Inception V3, ResNet34, MobileNet V1, and Xception achieved the best accuracies, despite having very different computational footprints.
2. **Training time was also inconsistent with model complexity.** Some lightweight models (e.g., SqueezeNet) trained slower than expected, while certain heavier models (e.g., Inception V3, Xception) trained efficiently.

STUDY LIMITATIONS

- **Fixed Hyperparameters.** All models were trained using identical hyperparameters. This design isolates architectural effects but likely underestimates the true performance of models that require tailored tuning (e.g., SqueezeNet, EfficientNet).
- **Hardware Specificity.** Results reflect performance on NVIDIA GPUs. Latency and efficiency trends may differ on CPU-only or low-power edge devices where memory patterns dominate.
- **Intrinsic Architectural Constraints.** Some architectures carry structural limitations that hinder convergence under standardized training settings. Examples include SqueezeNet's aggressive compression, DenseNet's feature-map accumulation, and the depth of large ResNet variants. These models may require architecture-specific schedules to reach full potential.

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